VISCAN AI ENGINEER

Take-Home Technical Assignment

Candidate technical assessment • Synthetic data only

Estimated effort	Submission window	Level
6-8 hours	3 calendar days	AI Engineer

Challenge Overview

Build a small proof of concept for a voice-first dental assistant that can locate patient reports and convert spoken clinical observations into a structured draft report. We are evaluating practical AI engineering, not mobile UI polish.

1. Required Features

A. Patient report retrieval (mandatory)

Given a typed or spoken request such as *"Open Priya's latest OPG report"*, extract the intent and patient name, query mock patient/report data, and return the correct report. Handle ambiguous patient matches and missing reports safely.

B. Clinical dictation to draft report (mandatory)

Accept audio or a speech-to-text transcript, e.g. *"Lower right third molar is horizontally impacted"*, and produce editable structured JSON with findings, anatomical site, and a readable draft report. Do not invent observations; label uncertainty and require clinician review.

C. Conversational follow-up (mandatory)

Support one context-dependent follow-up such as *"Summarize her report"* without asking for the patient again when the reference is unambiguous. Show the source report or evidence used for each answer.

2. Optional Bonus Feature (Choose One)

- **Voice/typed scan-booking agent:** Checks synthetic appointment slots and asks for confirmation before booking.
- **Report-aware chatbot:** Answers grounded questions about the selected report.

Note: Bonus work is not required for a strong submission.

3. Technical Expectations

- **Tech Stack:** Use any reasonable stack (Python/FastAPI, Node.js, etc.) and any LLM/STT provider; a simple web UI, CLI, or API demo is acceptable.
- **Mock Data:** Provide a small mock dataset (at least 5 patients, including 2 with similar names; at least 2 reports per selected patient). Use no real patient information.

- **Architecture & Tool Calling:** Show function/tool calling or equivalent deterministic API integration; separate LLM interpretation from business-rule validation.
- **Production Considerations:** Include basic failure handling, input validation, and a brief note on privacy, authorization, latency, and model/tool costs.
- **Testing:** Include at least 3 automated tests covering correct retrieval, ambiguous patient identity, and unsupported clinical claims.

4. Deliverables

- **Source Code:** Git repository or ZIP with runnable source code and mock data.
- **Documentation:** README with setup instructions, model/provider choices, environment variable sample, and known limitations.
- **Demo:** Short demo video (3-5 minutes) or a documented sequence of API calls with sample outputs.
- **Architecture Document:** One-page architecture explanation, including the voice/STT, LLM, tool/API, and report-generation flow.

5. Submission Instructions

Send the repository/ZIP link and demo evidence to the HR contact who shared this assignment. Include a short note describing what works, what remains incomplete, and what you would improve with more time.

Important Notes

- External paid APIs are optional. Mocking STT or using a typed transcript is acceptable if the architecture clearly supports voice input.
- Avoid spending time on real PACS integration, training a dental imaging model, production deployment, or mobile app packaging for this exercise.